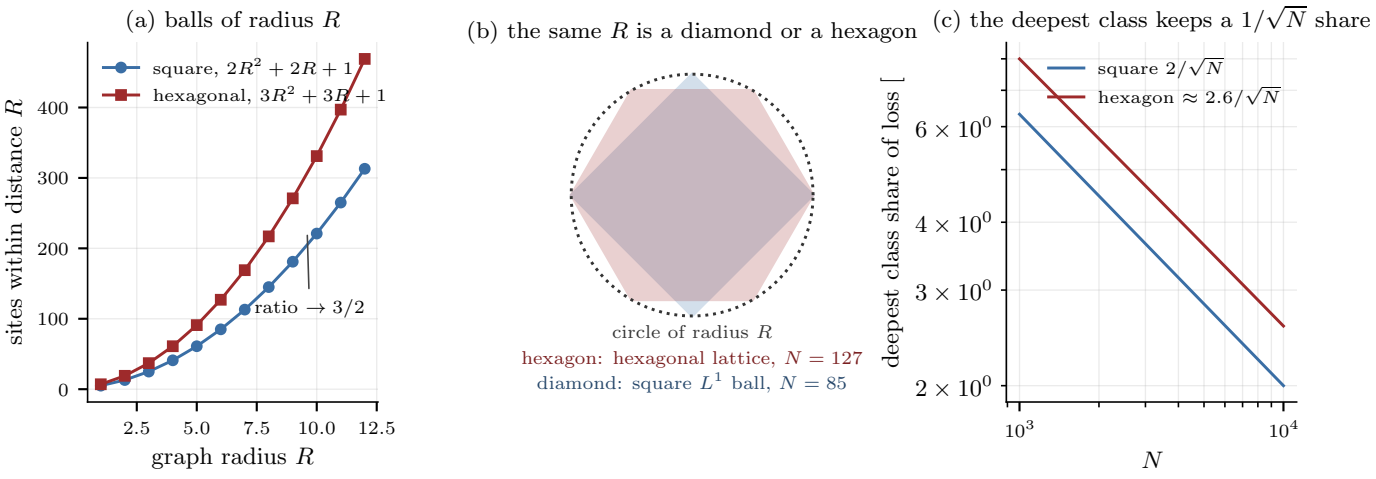

(a) balls of radius R
square, $2R^2 + 2R + 1$
hexagonal, $3R^2 + 3R + 1$
ratio $\to 3/2$
sites within distance R
graph radius R
0
100
200
300
400
2.5
5.0
7.5
10.0
12.5
(b) the same R is a diamond or a hexagon
circle of radius R
hexagon: hexagonal lattice, $N = 127$
diamond: square L^1 ball, $N = 85$
(c) the deepest class keeps a $1/\sqrt{N}$ share
square $2/\sqrt{N}$
hexagon $\approx 2.6/\sqrt{N}$
deepest class share of loss [
6×10^0
4×10^0
3×10^0
2×10^0
10^3
10^4
N